MC-12 microcentrifuge user manual

Fictional document for Northfield Lab Supply, a fictional company created for a portfolio project. Not for real use.

Item	Value
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Product	Benchtop Microcentrifuge MC-12
Applies to	NLS-MC12-STD, NLS-MC12-PLS
Revision	R1
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Status	current

1. Introduction

The MC-12 is a compact benchtop microcentrifuge for 1.5 ml and 2.0 ml tubes. It is sold in two versions: the Standard (fixed speed settings) and the Plus (variable speed, digital display and a second rotor).

Differences between the versions:

	Standard	Plus
Speed	3 fixed settings: 4,000, 8,000, 13,000 rpm	Variable 500 to 15,000 rpm in 100 rpm steps
Maximum RCF	14,200 ×g	18,900 ×g
Rotors supplied	12-place (1.5/2.0 ml tubes)	12-place (1.5/2.0 ml tubes) and 8-place PCR strip rotor
Display and timer	None; run continuously or short-spin	Digital speed and timer display; 10 s to 99 min or continuous
Warranty	1 year	2 years

2. Safety

- Place the unit on a level, stable surface with at least 10 cm clearance on all sides.
- Never run the centrifuge with an unbalanced load or with the lid open. The lid lock and imbalance cut-out are safety devices, not operating controls.
- Do not use tubes that are cracked, deformed or rated below the maximum RCF of your version.
- Do not spin flammable, explosive or hazardous biological material unless you have assessed the risk.
- Unplug the unit before cleaning or moving it.

3. Setting up

1. Remove the packaging and check the contents against the packing list: centrifuge unit, rotor(s), mains cable (UK plug) and this manual.
2. Place the unit on a level surface and connect the mains cable (230 V AC, 50 Hz).
3. Fit the rotor by pressing it onto the drive spindle until it seats, then tighten the rotor nut by hand.
4. Run the unit empty at the lowest speed for 10 seconds to confirm it runs smoothly.

4. Operation

1. Load an even number of tubes in opposite positions. For an odd number, balance with a tube of the same type and the same volume of water.

2. Close the lid until it latches.
3. Standard: choose one of the three speed settings, then press Start to run continuously, or hold Short spin for a brief run. Plus: set the speed and the time on the display, then press Start.
4. Wait until the rotor has stopped completely before opening the lid. Braking takes about 15 seconds.
5. Remove the tubes carefully and switch off at the mains.

5. Cleaning and maintenance

- Wipe the chamber and rotor with a damp cloth and mild detergent after use. Dry before closing the lid.
- Do not autoclave the rotor. Inspect it monthly for cracks or corrosion and replace it if damaged.
- Clean spills immediately; disinfect with 70 % ethanol if biological material has spilled.

6. Troubleshooting

Problem	Likely cause	What to do
Does not start	Lid not latched; mains cable not connected	Close the lid until it clicks; check the cable and socket
Loud vibration	Unbalanced tubes	Stop the run, check the tubes are balanced and opposite each other
Stops early or shows an error (Plus)	Imbalance cut-out triggered	Rebalance the load and restart
Rotor does not seat	Spindle dirty	Clean the spindle and rotor bore and refit

7. Warranty and contact

Warranty: Standard: 1 year. Plus: 2 years.

Warranty claims: contact Customer service with the order number and a description of the fault. The warranty covers manufacturing defects and does not cover damage from misuse, spills or unauthorised repair.

Contact: customerservice@northfieldlabsupply.example